rag -- 2026-10-28

5 new paper(s) · topic: Retrieval-augmented generation: retrieval quality, grounding and faithfulness, evaluation benchmarks, and RAG system design.

Worth reading in full

[P5] Tessera: Structure-Completed Evidence Units for Web Retrieval-Augmented Generation -- Tessera demonstrates a simple, training-free modification that consistently boosts accuracy across multiple benchmarks and retriever variants, directly linking improved evidence structuring to better grounding and offering immediate practical impact for RAG system design.

Themes

Several papers address how to improve the *quality* and *efficiency* of the retrieved evidence that feeds a generator. SpectRAG introduces a weighted chunk graph with spectral diffusion to select a token-budgeted, structurally diverse set of passages, achieving the highest overall recall and modest Exact-Match gains on MultiHopRAG [P2]. Tessera takes a different angle by redesigning the *unit* of evidence: it rebuilds Wikipedia pages into fine-grained, structure-completed units and re-ranks them without any extra training, adding 5-6 accuracy points across three QA benchmarks and outperforming fourteen alternative methods under matched token budgets [P5]. GEMS-RAG extends this idea to multimodal engineering standards, constructing a graph that links text, figures, and tables and routing visual retrieval only when needed, which improves retrieval effectiveness and QA on the MUTCD dataset [P4]. In contrast, the cost-aware adaptive policy of P1 shows that invoking a more expensive hybrid retriever on a subset of queries can match evidence hit rates of always-hybrid retrieval but does not yield statistically reliable improvements in answer correctness or faithfulness, highlighting a gap between retrieval metrics and downstream generation quality.

The papers also converge on the challenge of *grounding* and *faithfulness* evaluation. P1 evaluates answer correctness and faithfulness with provisional LLM judges and finds no clear advantage for the adaptive policy, while P4 reports evaluator-confidence scores from a secondary model but notes the lack of statistical confidence intervals. Tessera directly measures evidence delivery (from 72.1 % to 81.9 % of questions) and shows that better-structured evidence translates into higher accuracy, suggesting that fine-grained grounding can bridge the retrieval-generation gap. SpectRAG's limitations acknowledge that richer retrieval does not always produce proportional answer gains, especially for tasks requiring precise temporal ordering or abstention, echoing P1's finding that higher Hit@10 does not guarantee better downstream metrics.

Design choices for RAG systems differ markedly. P1 proposes a *two-action policy* that decides between keeping baseline evidence or invoking a hybrid lexical-semantic retriever based on a linear model trained on pre-retrieval features. SpectRAG builds a *graph-based selection* pipeline that balances relevance, proximity, and redundancy under a token budget. GEMS-RAG adds a *query-adaptive router* to decide when visual evidence is needed, while Tessera remains *training-free* and can be dropped into existing pipelines without changing the underlying retriever or reader. These varied architectures illustrate a spectrum from lightweight, plug-in enhancements (Tessera) to more complex, adaptive routing (P1, GEMS-RAG) and graph-centric selection (SpectRAG).

3 theme sentence(s) checked against the paper summaries.

At a glance

[P1] When Is Additional Retrieval Worthwhile? A Cost-Aware Study of Evidence Gain and Answer Reliability in RAG · relevance 0.53
A cost-aware adaptive retrieval policy can keep evidence hit rates comparable to always using a hybrid retriever while invoking the expensive hybrid step on only about a quarter of questions, but it does not yield statistically reliable gains in answer correctness or faithfulness.

[P2] SpectRAG: Weighted Chunk Graphs for Token-Budgeted Context Selection in Multi-Hop Retrieval-Augmented Generation · relevance 0.46
SpectRAG augments dense retrieval with a weighted chunk graph and spectral diffusion to select token-budgeted, structurally diverse context, improving multi-hop evidence recovery and modestly boosting answer accuracy.

[P3] LM-DSB: A Lightweight Multi-agent System for Automated Daily Safety Bulletin Generation in Construction · relevance 0.44
LM-DSB is a lightweight multi-agent system that automates daily safety bulletin generation on consumer-grade PCs, achieving near-human quality scores but still limited by knowledge scope and lack of field testing.

[P4] GEMS-RAG: Graph-grounded Engineering Multimodal Standards Retrieval-Augmented Generation · relevance 0.41
GEMS-RAG is a graph-grounded, multimodal retrieval-augmented generation system for engineering standards that integrates text, figures, and document structure, showing improved retrieval and QA performance on the MUTCD but with incomplete evaluation metrics.

[P5] Tessera: Structure-Completed Evidence Units for Web Retrieval-Augmented Generation · relevance 0.41
Tessera is a training-free evidence delivery layer that rebuilds Wikipedia pages into fine-grained, structure-completed units, boosting RAG answer accuracy by 5-6 points across text and screenshot pipelines without changing the underlying index, retriever, or reader.

[P1] When Is Additional Retrieval Worthwhile? A Cost-Aware Study of Evidence Gain and Answer Reliability in RAG

Zhouting Zhao · arXiv 2610.37358v1 · https://arxiv.org/abs/2610.37358v1 · relevance 0.53

TL;DR: A cost-aware adaptive retrieval policy can keep evidence hit rates comparable to always using a hybrid retriever while invoking the expensive hybrid step on only about a quarter of questions, but it does not yield statistically reliable gains in answer correctness or faithfulness.

Problem
Retrieval-augmented generation (RAG) often uses a fixed retrieval strategy, yet additional, more sophisticated retrieval may be unnecessary for many queries and improvements in evidence recall do not necessarily translate into more reliable answers [1][2]. The paper asks whether the incremental benefit of an extra retrieval step can be predicted from baseline retrieval signals and whether such a cost-aware decision can improve downstream answer quality [3][4].

Method
The authors design a two-action policy that, after an initial dense-retrieval plus cross-encoder reranking, either **Keep** the baseline evidence or invoke a **Hybrid** lexical-semantic retriever. A regularized linear model predicts the question-specific incremental evidence-gain using features available before routing; the model and decision threshold are selected via document-grouped cross-validation and then frozen for held-out evaluation [1][3]. Evaluation is performed on a held-out set of 60 QASPER articles (188 questions, 169 evidence-evaluable) [1][5].

Key results
- The adaptive policy triggers hybrid retrieval on 46 of 188 questions (24.5 %) [1][6].
- Evidence Hit@10 under the adaptive policy matches always-hybrid retrieval (78.70 %) and exceeds always-keep (76.92 %) [1][6].
- Unitized utility (Hit@10 minus 0.02 × hybrid-call indicator) is 0.7819 for the adaptive policy, compared with 0.7692 for always-keep and 0.7670 for always-hybrid [6].
- Generation-level provisional scores (0-2 scale) show no clear advantage: correctness 1.127 (adaptive) vs 1.133 (always-hybrid) vs 1.127 (always-keep); faithfulness 1.820 (adaptive) vs 1.813 (always-hybrid) vs 1.844 (always-keep) [6].

- Bootstrap intervals for utility and for correctness differences include zero, indicating no statistically reliable improvement [6][7].

Limitations
- Answer-level assessments rely on an LLM judge (same model family as the generator) with only partial manual review; no independent human validation is provided [5][7][8].
- Faithfulness scores exclude pure abstentions, limiting interpretation of overall grounding performance [7].
- The held-out evaluation uses only the QASPER training split; the official test set and other benchmarks are untested, restricting generalizability [7].
- Reported cost metrics are abstract unitized penalties, not real-time latency or token-cost measurements [7].
- The small number of rescue and harm instances and the simple ridge regressor may weaken policy learning and prediction reliability [7].

Sources

[1] arXiv 2610.37358v1.pdf, arXiv abstract

[2] arXiv 2610.37358v1.pdf, page 1

[3] arXiv 2610.37358v1.pdf, page 2

[4] arXiv 2610.37358v1.pdf, page 3

[5] arXiv 2610.37358v1.pdf, page 5

[6] arXiv 2610.37358v1.pdf, page 6

[7] arXiv 2610.37358v1.pdf, page 7

[8] arXiv 2610.37358v1.pdf, page 8

[P2] SpectRAG: Weighted Chunk Graphs for Token-Budgeted Context Selection in Multi-Hop Retrieval-Augmented Generation

Zafer Sahinoglu, Abdulazeez Mohammed, Buse Yaren Kazangirler, Cahit Berkay Kazangirler · arXiv 2610.36991v1 · https://arxiv.org/abs/2610.36991v1 · relevance 0.46

TL;DR: SpectRAG augments dense retrieval with a weighted chunk graph and spectral diffusion to select token-budgeted, structurally diverse context, improving multi-hop evidence recovery and modestly boosting answer accuracy.

Problem
Retrieval-augmented generation suffers from redundant top-k dense retrieval, loss of bridge evidence needed for multi-hop reasoning, and blindness to document structure such as adjacency, hierarchy, entities, and citations [2]. These issues limit the amount and diversity of evidence that can fit within the generator's context window [2].

Method
SpectRAG first seeds retrieval with a hybrid of dense FAISS scores, BM25, and utility-question matching, then expands the seed set over a weighted chunk graph whose edges encode semantic similarity, adjacency, hierarchy, entity overlap, citation links, and utility-question similarity [3][4]. The graph's Laplacian provides spectral coordinates that are diffused to propagate relevance (spectral diffusion) [4]. Finally, a token-budgeted, cluster-aware selection algorithm chooses chunks that balance relevance, proximity, cluster coverage, document coverage, and redundancy under a fixed token limit [4].

Key results
Experiments on MultiHopRAG show that SpectRAG attains the highest overall recall and Recall@10, while hybrid retrieval attains the best Recall@2, Recall@5, and MRR [6]. In generation, SpectRAG achieves the highest Exact Match, F1, and answer-containment scores among compared systems, though the margins

are modest [6]. The paper does not provide the exact numeric values for these metrics in the excerpts [5][6].

Limitations
The authors note that improved retrieval does not always translate into proportional gains in generated answers, especially for tasks requiring precise temporal ordering, direct comparison, or abstention, and that performance is sensitive to signal weighting, chunk size, and task type [7]. Additionally, the reported results lack detailed numbers in the provided excerpts [5].

Sources

[2] arXiv 2610.36991v1.pdf, page 3

[3] arXiv 2610.36991v1.pdf, page 7

[4] arXiv 2610.36991v1.pdf, page 11

[5] arXiv 2610.36991v1.pdf, page 12

[6] arXiv 2610.36991v1.pdf, page 20

[7] arXiv 2610.36991v1.pdf, page 23

[P3] LM-DSB: A Lightweight Multi-agent System for Automated Daily Safety Bulletin Generation in Construction

Xianxiang Zhao, Advik Mehta, Falak Sethi, Brian Gue et al. · arXiv 2610.36433v1 · https://arxiv.org/abs/2610.36433v1 · relevance 0.44

TL;DR: LM-DSB is a lightweight multi-agent system that automates daily safety bulletin generation on consumer-grade PCs, achieving near-human quality scores but still limited by knowledge scope and lack of field testing.

Problem
Construction sites still rely on manually created daily safety bulletins (DSBs), a process that consumes 2-3 hours each day and yields inconsistent quality, with 25 % of crews never performing pre-task planning (Olson et al., 2016)[1]. This manual burden diverts safety personnel from on-site supervision and hampers timely communication of site-specific hazards[1].

Method
The study proposes LM-DSB, a system that combines retrieval-augmented generation (RAG) with a locally deployable Llama 3-8B model, fine-tuned on construction safety documentation and grounded in 37 HSE documents[1]. LM-DSB uses a hierarchical multi-agent architecture--planner, researcher, and writer agents coordinated by a supervisor--implemented with the open-source LangGraph framework to manage context and task decomposition according to Job Hazard Analysis principles[3]. Fine-tuning employed LoRA on 500 domain samples with 4-bit quantization, batch size 2, learning rate 5e-5, and three training epochs[5].

Key results
In Phase 1, the multi-agent RAG configuration (Llama 3-8B) outperformed all alternative setups, achieving TF-IDF 0.70, BLEU 0.39, ROUGE-1 0.66, and a document length ratio of 1.01 versus 0.39 for a single-agent RAG system[1]. Phase 2 expert evaluation with 40 construction-management professionals yielded mean Likert scores ranging from 3.50 to 4.20 across five quality dimensions, with the highest scores for regulatory compliance (M = 4.20, SD = 0.65) and factual accuracy (M = 4.10, SD = 0.55)[6]. Completeness (M = 3.98), comprehensibility (M = 3.77), and actionability (M = 3.50) were lower but still acceptable[6]. The authors conclude that lightweight models (<=8 B parameters) can produce acceptable safety documentation when enhanced with multi-agent orchestration and domain-specific fine-tuning[7].

Limitations
The authors note that the system's knowledge base is limited in scope, language support is constrained,

and no field deployment validation has been performed[1]. They also acknowledge the absence of multimodal integration and the need for further workflow adoption studies[1].

Sources

[1] arXiv 2610.36433v1.pdf, arXiv abstract

[3] arXiv 2610.36433v1.pdf, page 12

[5] arXiv 2610.36433v1.pdf, page 23

[6] arXiv 2610.36433v1.pdf, page 29

[7] arXiv 2610.36433v1.pdf, page 34

[P4] GEMS-RAG: Graph-grounded Engineering Multimodal Standards Retrieval-Augmented Generation

Abdul Hannan Azad, Irvin Wang, Jiwan Jiang, Chenchen Kuai et al. · arXiv 2610.33277v1 · https://arxiv.org/abs/2610.33277v1 · relevance 0.41

TL;DR: GEMS-RAG is a graph-grounded, multimodal retrieval-augmented generation system for engineering standards that integrates text, figures, and document structure, showing improved retrieval and QA performance on the MUTCD but with incomplete evaluation metrics.

Problem
Engineering standards are difficult for large language models because they are hierarchical, multimodal, and densely linked through figures, tables, cross-references, and normative rules, and existing RAG methods that rely on unstructured text retrieval struggle with joint reasoning over these elements[1].

Method
GEMS-RAG builds a multimodal knowledge graph by hierarchically parsing documents, extracting fine-grained visual entities, and linking them to textual provisions and regulatory rules; it then performs graph-grounded hybrid retrieval that combines dense, sparse, and visual retrieval with graph expansion[1]. During inference a query-adaptive router decides whether visual evidence should be used (?(q)=1) based on deterministic rules such as explicit figure or table identifiers, thereby gating visual retrieval to avoid unnecessary context[2].

Key results
The paper reports an evaluator-confidence summary of 0.85 with a confidence interval of [0.78, 0.90] supplied by a secondary language-model auditor[4].
It states that graph-grounded multimodal retrieval improves retrieval effectiveness and question-answering performance compared with text-only, multimodal, and graph-based RAG baselines on the MUTCD, though the exact numeric gains are not provided in the excerpts[1].

Limitations
The repeatability component of the overall 100-point scoring framework is pending and therefore not included in the reported scores[6].
For the no-router ablation, confidence-related metrics (Brier score, ECE, calibration points, 12-point reliability score, and the full 97-point score) are undefined, limiting direct comparison[6].
The evaluator-confidence interval is a judgmental interval from a secondary model rather than a statistical confidence interval, which may affect the robustness of the reported confidence[4].

Sources

[1] arXiv 2610.33277v1.pdf, arXiv abstract

[2] arXiv 2610.33277v1.pdf, page 4

[4] arXiv 2610.33277v1.pdf, page 13

[6] arXiv 2610.33277v1.pdf, page 20

[P5] Tessera: Structure-Completed Evidence Units for Web Retrieval-Augmented Generation

Peichun Hua, He Sun, Zirui Zhuang, Qi Qi et al. · arXiv 2610.37080v1 · https://arxiv.org/abs/2610.37080v1 · relevance 0.41

TL;DR: Tessera is a training-free evidence delivery layer that rebuilds Wikipedia pages into fine-grained, structure-completed units, boosting RAG answer accuracy by 5-6 points across text and screenshot pipelines without changing the underlying index, retriever, or reader.

Problem
Web retrieval-augmented generation pipelines lose answer evidence when pages are split into fixed-size text chunks or screenshot tiles, especially for structured content such as tables and infoboxes, because the chunking separates these elements from their headers and context [1][7]. Tracing evidence from source HTML to the reader shows that a substantial portion of relevant evidence never reaches the model intact [1][7].

Method
Tessera introduces a universal, training-free evidence delivery layer that (i) rebuilds each retrieved page's HTML into fine-grained "evidence units" (block units for paragraphs/list items and row units for table rows), completing each unit with its page title, section path, and table headers [3]; (ii) selects the most relevant units via a lexical prefilter (top 100) followed by a cross-encoder re-ranking using the pipeline's existing reranker, adding no extra model [3]; and (iii) packs the selected units into the reader's context while keeping the token budget unchanged, allowing it to plug into text, screenshot, and agentic pipelines without retraining [1][2][4].

Key results
- Across three QA benchmarks (Natural Questions, NQ-Tables, SimpleQA; 7,851 questions total), Tessera adds 5-6 accuracy points to both text and screenshot pipelines [2].
- In multi-step search agents (FRAMES, MuSiQue), it yields up to a 5.2-point gain [2].
- On NQ-Tables, gains range from 4.4 to 6.3 points, with the largest improvements where baseline chunks miss evidence [5].
- Across six alternative retriever variants, Tessera contributes 4.9-5.5 points each; with the strongest retriever (Query2doc) the gain is 5.1 points, and the combined system reaches 76.7 % single-shot accuracy [5].
- For a larger reader (Gemma-4 31B), baseline improves by 1.3 points and Tessera adds 4.7 points (text) and 5.9 points (screenshots) [5].
- Compared to fourteen alternative methods under matched token budgets, Tessera outperforms them by 4.1-6.1 points, and a 3K-token Tessera context beats a 12K-token chunk or tile context [2][5].
- Evidence delivery rises from 72.1 % to 81.9 % of questions, with 13.8-point gains for questions whose evidence resides only in tables or infoboxes [6].

Limitations
The evaluation is limited to the August 2025 English Wikipedia snapshot; extending to the broader web will require handling layout tables, client-side rendering, and noisier markup [7]. Remaining errors stem from outdated references, grading mistakes, and failures of the lexical prefilter to retrieve needed units, which a semantic prefilter might address, but larger prefilters change accuracy by at most 0.3 points [6][7]. The layer also requires access to page HTML at query time.

Sources

[1] arXiv 2610.37080v1.pdf, arXiv abstract

[2] arXiv 2610.37080v1.pdf, page 2

[3] arXiv 2610.37080v1.pdf, page 4

[4] arXiv 2610.37080v1.pdf, page 5
[5] arXiv 2610.37080v1.pdf, page 6
[6] arXiv 2610.37080v1.pdf, page 8
[7] arXiv 2610.37080v1.pdf, page 9